

REAL-TIME EDGE-BASED SMART PATIENT CARE AND ALERT SYSTEM

PROBLEM STATEMENT:

Paralyzed, bedridden or old patients find it difficult to communicate simple needs like food, water, medicine or emergency assistance. Caregivers might not pick up the requests, and family members are not always able to check the patients. The solutions that are currently available are hospital tied, costly, or do not contain accountability or real-time feedback loops.

PROPOSED SOLUTION:

The proposed Smart Patient Assistance System based on an Edge-Powered system allows a patient to express their needs in the most effective way, provide accountability to the caregiver, and proactive help. This solution features multi-modal patient input, local edge processing, real-time alerts, pattern-based recommendations to provide an intelligent, privacy-performing, and reliable care system.

Key Features:

1. Multi-Modal Patient Input:

- Common requests (Food, Water, Medicine, Help, Light/Fan) using touch buttons in capacitive manner.
- Voice recognition in the EFR32MG26/SiWx917 through local recognition using TinyML and facilitates hands-free communication with patients who are not so mobile.

2. Edge-Based Processing & Prioritization:

- All AI inference and decision-making is done on the EFR32MG26/SiWx917 edge processor.
- The priority given to requests is on urgency (Emergency > Medicine > Food/Water).
- There is no cloud dependency guaranteeing low-latency, secure and privacy of operation.

3. Caregiver Acknowledgment & Family Monitoring:

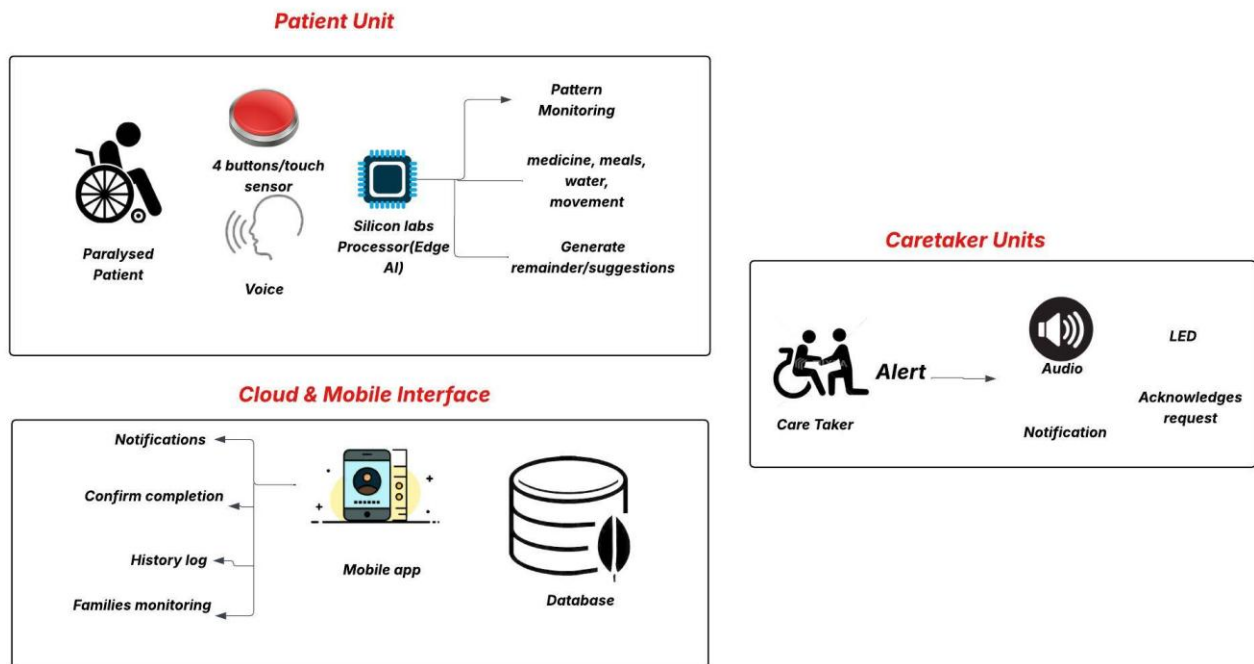
- Every patient request should be taken into consideration in the caregiver mobile app.
- In the event that a request is not validated within a specified period of time, caregivers receive reminders and family members may be contacted to provide accountability.
- Patients are provided with LED, audio or vibration feedback on the acknowledgement of their requests.

4. Pattern Monitoring & Proactive Assistance:

- The system becomes familiar with patient routines and habits e.g. meal times, medication schedules, water intake and move patterns.

- According to these patterns, proactive reminders are produced by the edge processor e.g., drinking water, patient needs bathroom support, or medication reminder.

BLOCK DIAGRAM:



HARDWARE COMPONENTS:

- **EFR32MG26/SiWx917** – Edge AI processor + sensor interface + wireless communication
- **Capacitive touch buttons** – 3–5 buttons for patient commands
- **MEMS/inbuilt microphone** – For TinyML voice recognition
- **LED / buzzer** – Feedback / acknowledgment to patient
- **Battery**

SOFTWARE COMPONENTS

- **TinyML model for voice command recognition** – Runs on EFR32MG26
- **Edge pattern monitoring software** – Logs patient activity, generates proactive suggestions
- **Priority-based request handling** – Determines urgency of patient requests

FEASIBILITY:

- Core system operates complete edge to low, private, and offline.
- Scalable architecture: begin with buttons + voice input;
- Effective and sensible: enhances patient autonomy, caregiver responsibility and avoids inattentive care.